

Cybercrime

1. **Real-Time AI-Based Phishing Email Detection System:** Design a system that uses AI to analyze email headers, textual patterns, metadata, and attachments to detect and quarantine phishing emails in real time.
Deliverables: Trained phishing detection model, API for email integration, dashboard for alerts and logs.
2. **Intelligent Financial Fraud Detection using UPI and VoIP Data:** Develop a smart system that uses machine learning to detect and alert about fraudulent UPI transactions, scam VoIP calls (including spoofed numbers), and suspicious payment patterns.
Deliverables: Dataset simulation, fraud scoring engine, alert system prototype for real-time monitoring.
3. **AI-Based Social Media Monitoring Tool for Crime Trends:** Create an AI solution that scans public social media content to detect hate speech, cyberbullying, fake accounts, incitement to violence, and threats to public safety.
Deliverables: Dashboard with analytics, real-time alert system, NLP-based threat categorization engine.
4. **Ransomware Incident Simulation & Training Platform:** Develop an interactive simulator for law enforcement officers to understand ransomware attack scenarios and build strategic response mechanisms.
Deliverables: Web-based simulation interface, ransomware propagation logic, and scenario-building toolkit.
5. **Deepfake & AI-Generated Media Detection Tool:** Design a forensic tool to identify manipulated images, videos, or voice recordings used for impersonation or blackmail.
Deliverables: Deep learning model, test suite of fake vs. original content, browser-based interface for testing.

Smart Policing

6. **Facial Recognition System for Identifying Missing Persons:** Build a tool that compares CCTV footage, social media photos, and public databases to help trace missing individuals using facial recognition.
Deliverables: Face detection model, matching engine, interface for uploading and viewing results.
7. **Voice Emotion Recognition from Emergency Calls:** Develop an AI system that can analyze voice inputs from helpline/emergency calls to detect distress, aggression, or urgency using tone analysis and NLP.
Deliverables: Real-time audio analyzer, pre-trained emotion classification model, dashboard for call responders.
8. **Voice & Chatbot Assistant for Police Personnel:** Create a mobile-friendly, multilingual chatbot that provides instant access to legal sections (IPC, CrPC), departmental forms, and SOPs.
Deliverables: Chatbot application, IPC/CrPC searchable database, voice-to-text integration.
9. **Automated Traffic Violation Detection System Using AI:** Design a smart traffic monitoring system that detects and records violations like speeding, signal jumping, and helmet-less driving using live CCTV feeds.
Deliverables: Video feed parser, violation classification model, violation reporting module.
10. **Suspicious Behavior Detection System for Public Surveillance:** Develop a computer vision system that identifies abnormal/suspicious movements in real-time (e.g., loitering, unattended baggage, sudden dispersals).

Deliverables: Pre-trained object detection model, alert module, real-time video annotation system.

Cyberthon.ai 2025

Dark Web & Financial Crimes

11. **Dark Web Activity Monitoring and Alert System:** Design a system to crawl dark web marketplaces and forums to extract and analyze discussions or listings involving drugs, arms, stolen data, or fake IDs relevant to India.
Deliverables: Custom crawler, keyword alert engine, categorized threat intelligence dashboard.
12. **Cryptocurrency Transaction Tracker for Illicit Activities:** Build a visual analytics tool to trace cryptocurrency wallets, transactions, and movement across exchanges to flag suspicious activity related to ransomware or dark web trade.
Deliverables: Wallet tracing engine, blockchain visualizer, wallet-risk scoring algorithm.
13. **AI-Driven Dark Web Crawler and Alert Generator:** Create an AI-enabled tool to index and monitor onion sites, marketplaces, and forums for leaked Indian data or criminal content.
Deliverables: Crawler with NLP filtering, alert system with categorization, visual search interface.

AI in Forensics & Investigation

14. **Audio/Video Enhancement System for Forensic Evidence:** Develop a tool that enhances low-quality audio/video inputs for clarity in forensic investigations (e.g., noisy calls, grainy CCTV).
Deliverables: AI model for de-noising/enhancement, comparison module, forensic-ready export formats.
15. **Crime Scene Image Analysis & Reconstruction Assistant:** Design an AI tool that uses uploaded crime scene images to reconstruct the sequence of events or generate potential 3D layouts.
Deliverables: Image parser, event modeling engine, 2D/3D layout generator with annotation features.

Additional Proposed Problem Statements (Relevant & Feasible)

16. **VoIP Scam Call Pattern Detection & Blacklist Engine:** Create a tool that analyzes patterns in incoming scam calls (e.g., lottery frauds, KYC update frauds) and blocks recurring VoIP abuse.
Deliverables: VoIP dataset simulation, scam classifier engine, integration with call-blocking app.
17. **UPI Scam Detection and Victim Alert System:** Develop an ML-based system that learns from UPI fraud complaint databases and flags suspicious transaction behavior in real-time to help warn victims.
Deliverables: Transaction analysis model, fraud alert generator, reporting interface for victims.
18. **SIM-Swap Fraud Early Detection Assistant:** Design a system that can detect early signs of SIM-swap fraud by correlating sudden changes in network behavior, UPI deregistrations, and OTP bypass logs.
Deliverables: Telco + bank signal simulator, fraud correlation engine, alert dashboard.

Expected Hackathon Deliverables for All Problem Statements

- Working prototype or MVP (Minimum Viable Product)
- Pitch deck explaining problem-solution fit, user flow, tech stack
-  debase with documentation (Share GitHub repo URL)
- Demo video (3–5 minutes max) Or presentation as per template

- Presentation to jury with use-case relevance to Chandigarh Police and Master-Hacks
- Optional: Dataset, scalability roadmap, or plan for pilot deployment (future roadmap)